

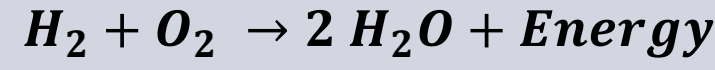
Toward Climate-Optimal Trajectory Optimization for H2-powered Aircraft

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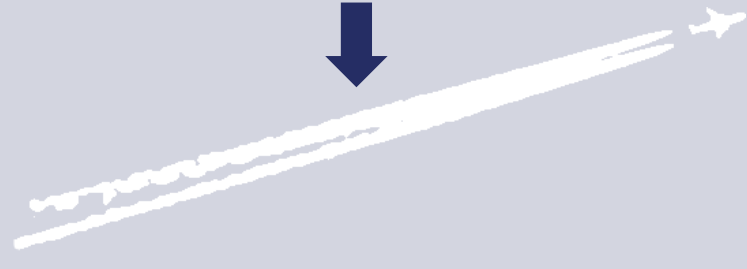
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Motivation

H₂ propulsion is a promising solution to mitigate aviation climate impact



Eliminates **direct CO₂ emissions**
Produced by renewable sources



What about **non-CO₂**?
Rerouting is also possible!

While the application of trajectory planning is being studied for the conventional kerosene aircraft [1], literature for the future H₂ fleet is completely missing.

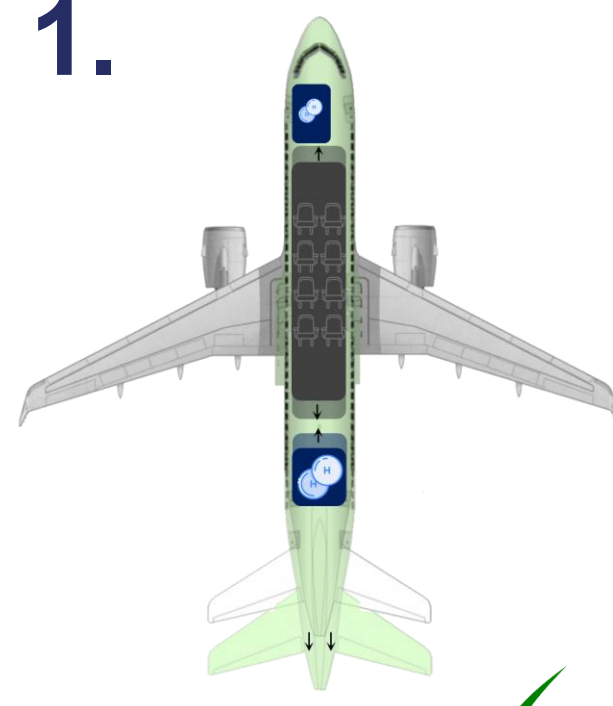
[1] Simorgh, Abolfazl, et al. "A comprehensive survey on climate optimal aircraft trajectory planning." Aerospace 9.3 (2022)

It is needed to :

- Understand its **capabilities** and **most beneficial operation** condition
- Quantify the extent of these benefits and its derived **costs**
- Foster and ease its **integration in the ATM** system
- Perform effective **climate trajectory optimization**

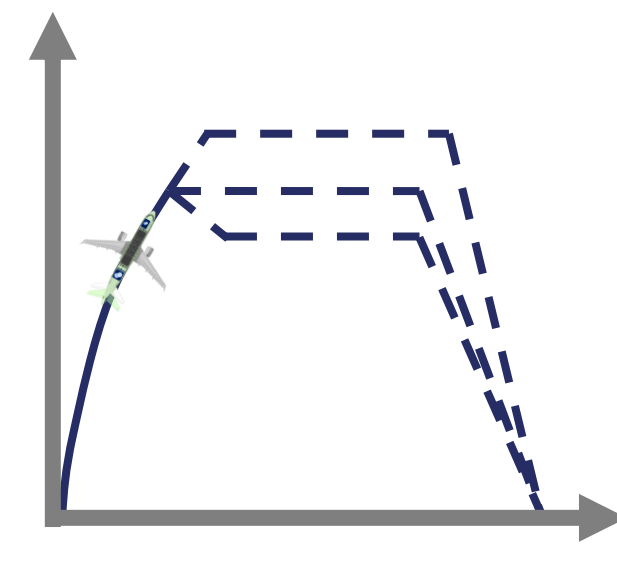
This work advances on the **development of tools and studies** on H₂ aircraft trajectory optimization, also considering climate impact!

1.



Design ✓

2.



Trajectory

3.



Integration

H₂-powered evolution from A320

- Elongated fuselage to allocate cryogenic tanks
- Tailored H₂ Turbofan

[2] R. Quiben Figueroa, R. Cavallaro, M. Soler, *Hydrogen Aircraft Multidisciplinary Optimization*, in preparation

Dynamical model

Aerodynamics

A **flexible parasitic drag model** is developed. It is calibrated against benchmark data from BADA, tailoring **wave drag contributions** (C_{Dw}) while accounting the geometrical modifications (C_{Df}) required by the evolved airframe.

skin-friction drag

$$C_{Df} = \frac{C_{fe} \cdot S_{wet}}{S_{ref}}$$

equivalent skin-friction drag

$$C_{fe} = 0.00258 + 0.00102 \cdot e^{-6.28 \cdot 10^{-9} \cdot Re_{Swet}} + 0.00295 \cdot e^{-2.01 \cdot 10^{-8} \cdot Re_{Swet}}$$

calibrated parasitic drag

$$C_{D0} = C_{Df} + C_{Dw} = k_0 \cdot C_{Df} + w_0 + w_1 M + w_2 M^2 + w_3 M^3 \dots$$

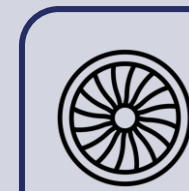
total drag

$$C_D = C_{D0} + k C_L^2$$

Propulsion

Polynomial fitted functions are generated from performance data of a **CFM56-equivalent H₂ turbofan**, designed and simulated using the pyCycle platform.

Mach
Altitude
Throttle



$$f(M, h, \delta) = f_0 + \sum_{i=1}^{20} k_i M^{p_{i1}} h^{p_{i2}} \delta^{p_{i3}}$$

Thrust
Fuel Flow

Application



ROC library for trajectory optimization in free-routing airspace is augmented with H₂ aircraft models

[3] Simorgh, A., Soler, M., Dietmuller, S., Matthes, S., Yamashita, H., Castino, F., Yin, F. (2024). *Robust 4D climate-optimal aircraft trajectory planning under weather-induced uncertainties: free-routing airspace*. Transportation Research Part D, 131.

Cost Optimal Flight

"Real-case" application comparing **4D trajectory optimization** performance for a flight between Tenerife and London. A parametric analysis of fuel- and time-cost scaling factors is conducted to assess their impact on the optimized trajectories

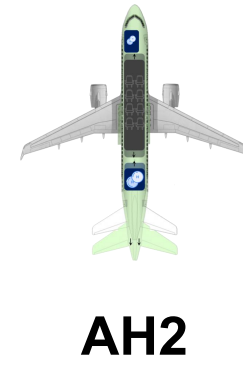


t_0 : 26/01/2024 06:00 UTC
Tenerife to London

$$f(x) = C_M \cdot \text{Fuel}(x) + C_T \cdot \text{Time}(x)$$



A320-214

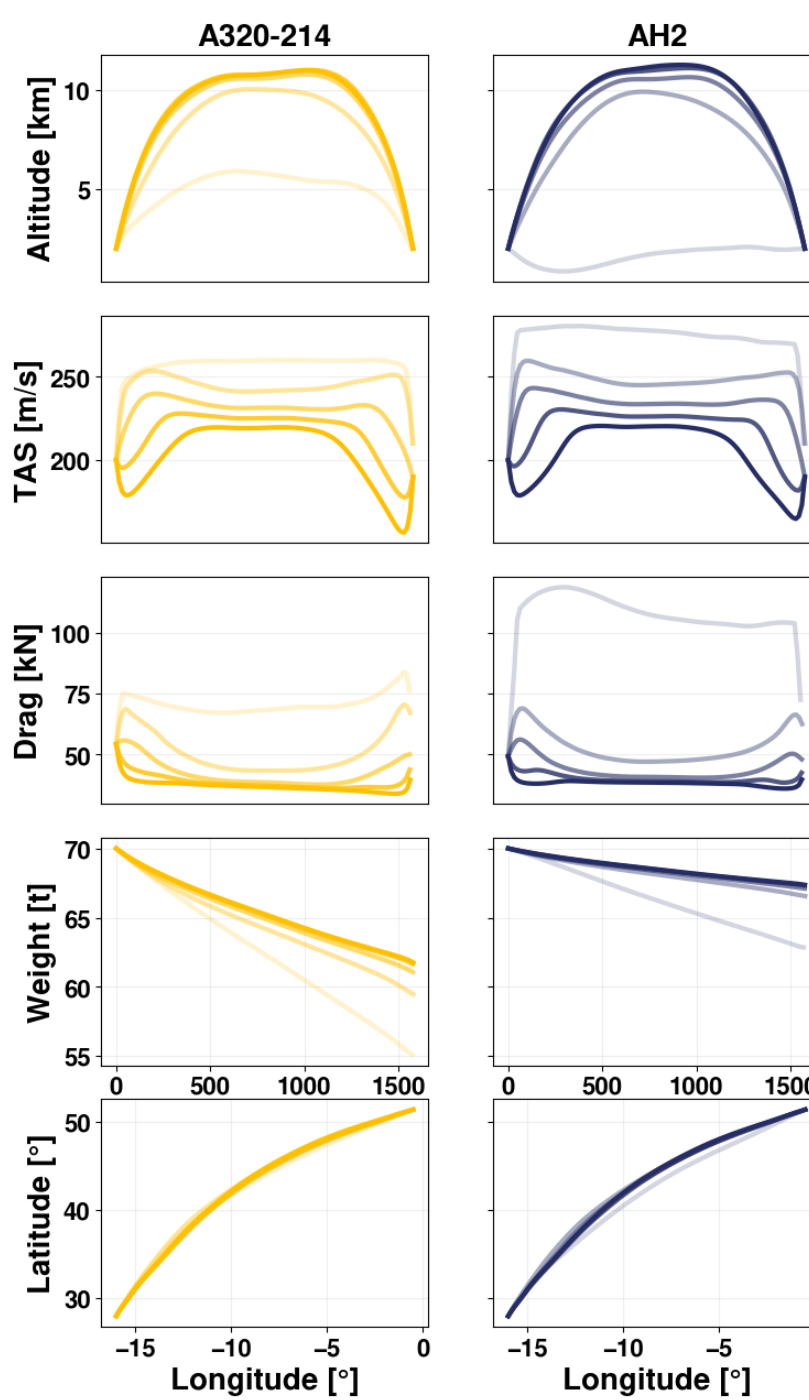


AH2

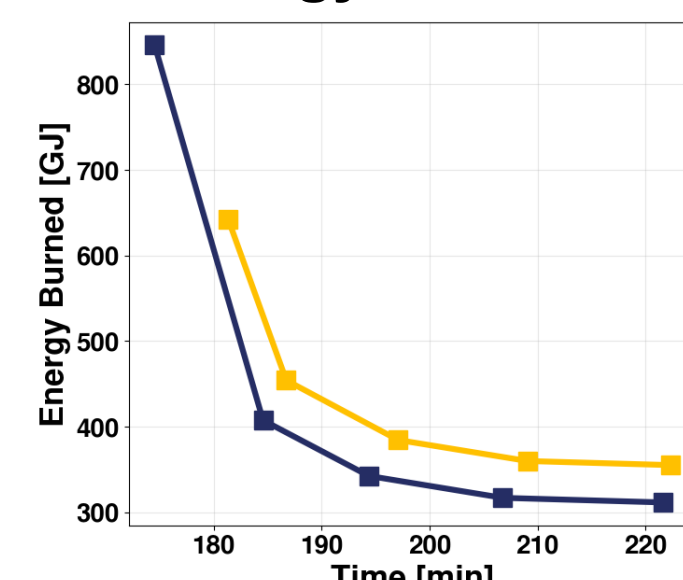
Cost weights

- $C_T=1.00, C_M=0.00$
- $C_T=0.75, C_M=0.25$
- $C_T=0.50, C_M=0.50$
- $C_T=0.25, C_M=0.75$
- $C_T=0.00, C_M=1.00$

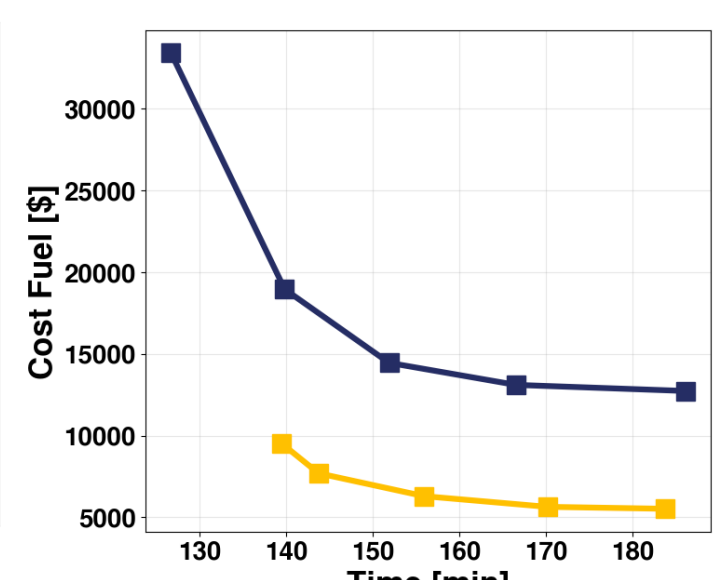
Trajectory Profiles



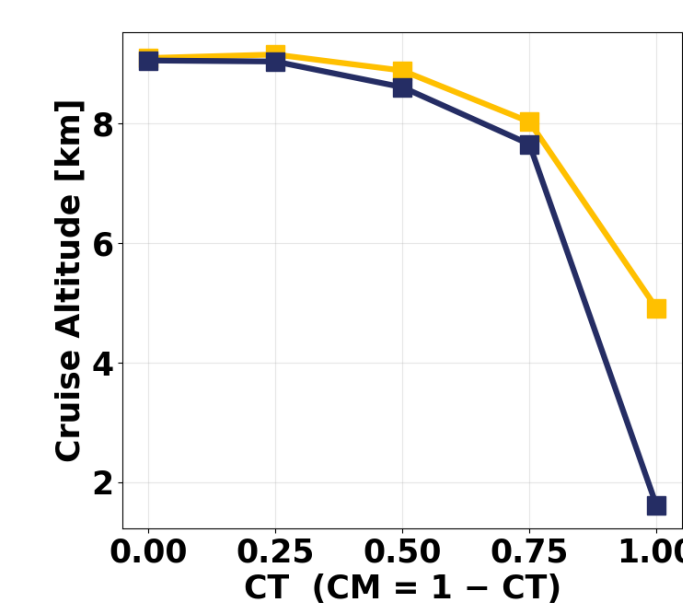
Energy vs Time



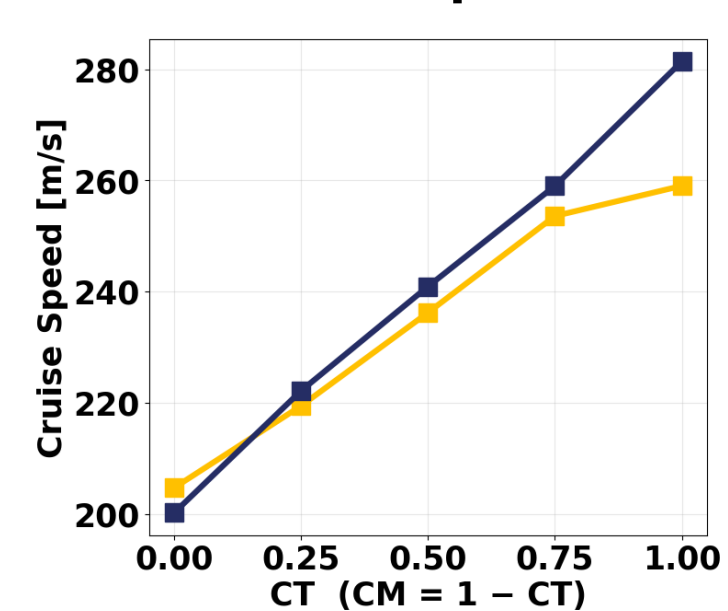
Cost vs Time



Mean Altitude



Mean Speed



Climate Optimal Flight

COCiP gridded model is used for estimating contrail energy forcing (EF), for JetA and H₂ fuels, over the airspace domain. The case presented before is now analyzed **with and without contrail impact**, included with the scale factor: environmental index (EI).

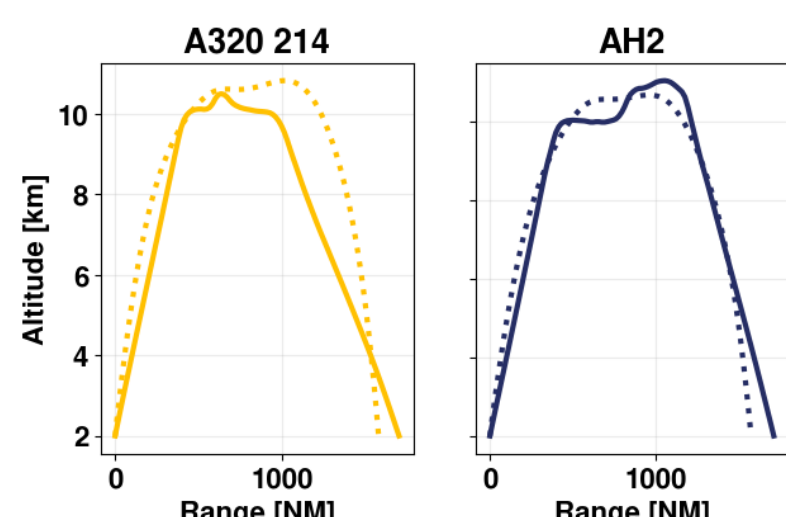
[4] Engberg, Z., Teoh, R., Abbott, T., Dean, T., Stettler, M.E.J., Shapiro, M.L. "Forecasting contrail climate forcing for flight planning and air traffic management applications: The CocipGrid model in pycontrails 0.51.0.

$$f(x) = C_I \cdot (C_M \cdot \text{Fuel}(x) + C_T \cdot \text{Time}(x)) + E_I \cdot \text{EF}(x)$$

$$C_I = 1.0 \quad C_M = 0.5$$

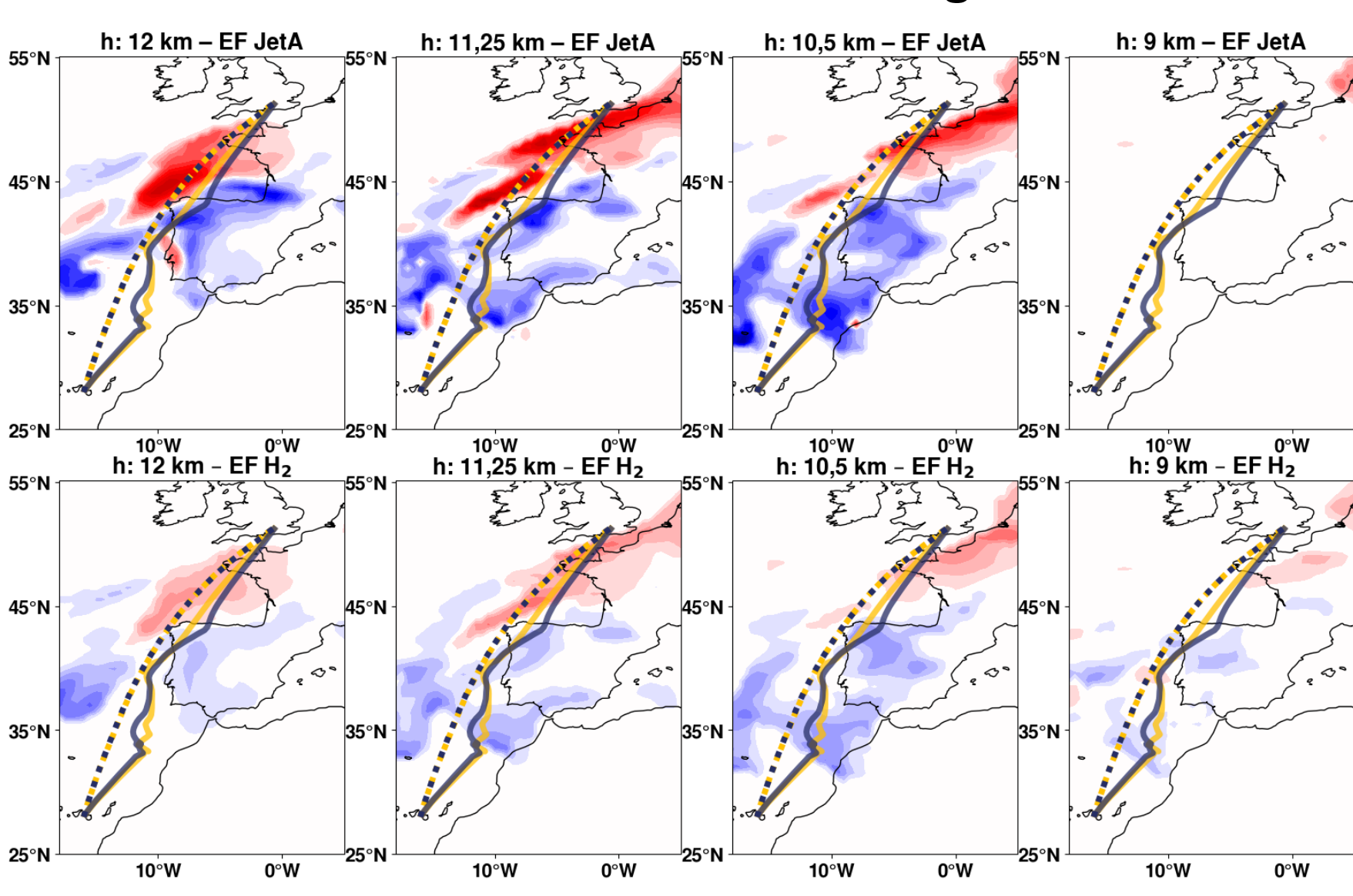
$$C_T = 0.5$$

Vertical Profiles

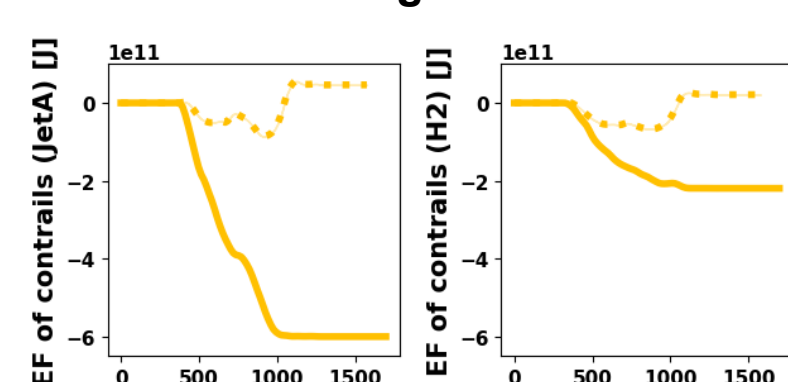


- A320 Cost-Opt (EI=0)
- A320 Climate-Opt (EI=1e-17)
- AH2 Cost-Opt (EI=0)
- AH2 Climate-Opt (EI=1e-17)

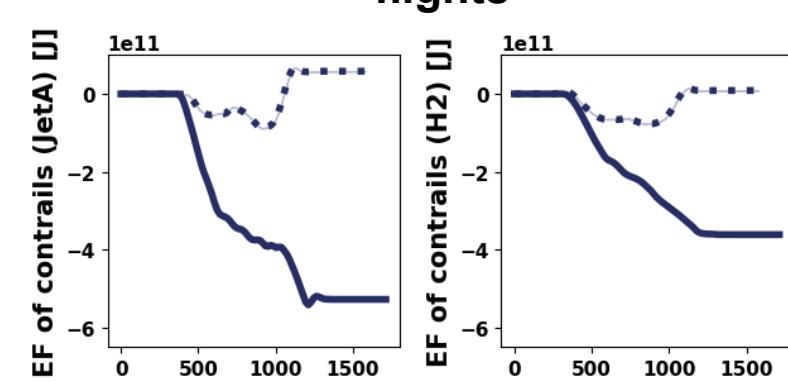
Lateral Profiles over EF regions



Accumulated EF of A320 flights



Accumulated EF of AH2 flights



Future Steps

Trajectory optimization could be a game changer for the future feasibility of H₂-powered aircraft. However, **major uncertainties remain** in technology performance and contrail formation. Further comprehensive parametric studies, i.e. covering **range, seasonality, diurnal conditions, and hydrogen fleet allocation**, are still mandatory and the most urgent next steps.